Supplemental methods and evidence

Consolidated supplement S1 to S4

Reading guide

This supplement explains construction and evaluation of the study population and measurements (S1), research-content comparisons (S2), funding connections and the exploratory opioid comparison (S3), and employment and the Research and Clinical Role Measurement Instrument (S4). Each analysis specifies its population, counting unit and uncertainty.

The repository catalog provides complete tables, figures and source notes for corpus and measurement (DEP1), research content (DEP2), funding and connections (DEP3), and employment and RC-RMI (DEP4). Data notes define table use; reproduction instructions distinguish saved-result inspection, aggregate arithmetic and re-estimation. The optional priority and objective–employment extensions are outside this distribution.

Supporting files, including the reader and scorer packages, accompany this supplement in the shared Zenodo record (10.5281/zenodo.23897503). Licensed source text, identifying linkage evidence and raw reference forms are excluded.

Navigate to S1 data and measurement, S2 research content, S3 funding and connections, S4 employment and role measurement, or References. Core tables and figures are numbered within each section; the source-to-final display crosswalk records their original labels.

S1 Data construction and measurement

S1.1 Corpus inclusion and counting units

We searched ProQuest Dissertations & Theses Global in August 2026 using indexed fields other than full text (NOFT). The saved query used publication dates 1 January 2010–31 December 2023; the roster used completion years 2010–2023. We retained English-language records from U.S. institutions, excluding PQDT UK & Ireland. Degree Date supplies completion year and Date publication year. The exact expression is search/topical_query.txt in the separate reader package; search/REVIEWER_SEARCH.md specifies rerun procedures.

The collection contained 32,578 distinct identifiers, including 28,137 English-language records. Geographic processing and deduplication yielded 21,073 roster records. Historical deduplication used identifiers, then normalized title–author–brief-citation-year keys, protecting titles shorter than eight normalized characters. Every roster identifier occurs in the collection, but stepwise historical removals could not be reconstructed. Database coverage and topical retrieval are distinct limitations; lexical queries may represent field facets unevenly (Zitt and Bassecoulard, 2006). Neither reference evaluation nor the award-linked check establishes overall search recall.

A record is a ProQuest identifier; a work is its underlying thesis or dissertation. Retrieval and coding use records; substantive output analyses use reconciled distinct works. Dissertation here includes master's theses. A work's degree does not establish its author's highest or terminal qualification. Different works by one author remain separate.

An aim is in scope when it substantively concerns a study-covered substance and addresses substance use, addiction, exposure, pharmacology or explicitly substance-related care, services, policy, education

or methods. Endogenous-system research is also in scope without a focal exogenous drug or an explicit substance-use, addiction or substance-related harm question. Background or sample mentions alone are insufficient; therapeutic use for unrelated conditions does not by itself establish inclusion. This operational definition does not encompass the entire mission of NIDA or NIAAA. The integrated revision-15 codebook defines the fields and rules.

Coding evidence was restricted to titles and abstracts, excluding full text and institution, degree, department, year and funding labels. There was no minimum abstract length. Models were instructed to preserve unsupported fields as uncertain, identify described aims when explicit aims were absent, and separate aims where scope or the substance–purpose connection differed. Software classified a record as in scope if any aim was in scope, undetermined if none was and at least one remained unresolved, and out of scope otherwise. Addiction and substance-use focus was recorded separately and did not gate inclusion. Table S1.1 accounts for the population.

Table S1.1 Population accounting

Stage or disposition	Records or works	Interpretation
Established input roster	21,073	Distinct ProQuest record identifiers after the search and geographic processing described in Section S1.1.
Assessable abstract	21,045	Records with assessable abstracts; this is not a count of completed model responses.
Not assessed	28	Eight blank abstracts and 20 recognized placeholders; no model request.
Adopted in-scope records	13,229	At least one aim meets the substance-related or endogenous-system inclusion rule.
Adopted out-of-scope records	7,702	No in-scope or unresolved aim.
Undetermined records	114	No in-scope aim and at least one unresolved aim.
Extra in-scope record identifiers consolidated into another scientific work	2	Two pairs of in-scope records each represent one adopted work.
Adopted in-scope dissertations and theses	13,227	Distinct scientific works after two extra identifiers are consolidated.

The 21,045 assessable records plus 28 not assessed give 21,073; 13,229 included identifiers map to 13,227 works. The human reference evaluates the substance-related criterion and its content targets. Systematic six-field role data exist for 133/200 reference records; reduced judgments for the remainder cannot support complete role-based derivation.

Table S1.2 Counting bases

Count	Basis
Candidate records	21,073 distinct ProQuest identifiers for retrieval and scope accounting; 21,045 have assessable abstracts and 28 are not assessed.
In-scope works	13,227 distinct theses and dissertations after reconciling 13,229 in-scope record identifiers.
All-topic works	Distinct theses and dissertations across all subjects, counted in the same institution–year–degree cells as the in-scope subset.
Institution–year–degree combinations	One institution, completion year and degree group per cell; works are aggregated within each cell for institutional comparisons (S3).
Investigator–work connections	Links from identified advisors or committee members to works. A work may have several links but counts once in work-level funding reach (S3).
Employment observations	Retained employment records for sampled authors; observation and role-scoreable bases differ when information is unavailable (S4).
Degrees awarded	IPEDS all-fields awards by institution, survey year and award level; each matched total is counted once.

These bases overlap rather than form successive exclusions. Award cores, award/person relationships, repeated award observations and matched works also differ. Coded values, cannot determine, structural nonapplicability and incomplete processing remain distinct. Each analysis states its denominator; measurement evaluation separates conditional agreement from correct delivery on all reference-coded values.

S1.2 Coding provenance and the reader workflow

Small, reused development comparisons favored GPT-5 for scope; GPT-4o was retained for content for budget reasons, without establishing content equivalence between models. Original production used frozen revision-15 prompts.

Table S1.3 Requested profiles for the reference evaluation

Stage	Requested model snapshot	Request settings	Output and role
Scope	GPT-5 2025-08-07	Low reasoning effort; max_completion_tokens 12000; temperature and seed unset	Scope units and fields; software derives inclusion
Routed content	GPT-4o 2024-08-06	Temperature 0; max_tokens 5000; seed unset	C1–C9 for in-scope targets only

Requests used the ProQuest TDM Studio Workbench OpenAI-compatible Chat Completions endpoint and structured JSON output, without a seed. Retained scope responses date from 2–10 October 2026 UTC, with failure/retry history into 11 October; original content responses date from 9–13 October. Additional responses on 14–15 October requested GPT-4o 2024-08-06, temperature 0 and max_tokens 8000. Requested settings do not establish the proxy's internal handling of reasoning effort. Table S1.3 specifies the reference-evaluation profiles.

The current reader package, version 0.3.0-rc2, provides a search-derived replication using the integrated revision-15 codebook. One scope pass assesses all six S1–S4 aim fields and record-level S5; software derives inclusion and fixes ordered target identifiers. Applicable substance fields are assessed during scope, with unsupported values retained as cannot determine. Later coding supplies C1–C9 only and cannot add aims, fill substance fields or alter scope or target membership. This describes the current reader workflow, not an exact reconstruction of historical execution.

For reproducibility, readers must supply their own authorized TDM Studio XML and Extended Metadata. Licensed-input processing and model calls are disabled by default. Indexing, metadata and model changes can alter future classifications; equal counts do not establish identical inputs. The reader README specifies licensing and operation. Human authors remain responsible for rules, reference coding, adjudication and interpretation; AI assistance does not supply human reference judgments (Geiger et al., 2021).

S1.3 Comparison frames and indexing

The all-topic comparison was retrieved separately using LA(English) AND SCHLOC("United States"), without the topical expression. Extended Metadata exports covered publication years 2009–2016 and 2017–2024, then retained completion years 2010–2023. No retained work was published after 2023. Nine doctoral works were published in 2009, completed in 2010 or 2011, and did not link to the topical roster. They remain in the comparison; the wider publication window accommodates differences between completion and publication dates and does not resolve recall of the topical query.

The frame used reviewed institutional histories and a 176-pair Degree/Degree Type crosswalk, with label-based fallback for unmatched pairs. It excluded unresolved institutions and specified post-closure records, and consolidated matching title–author records and related titles by the same author within institution, year and degree level. The broad frame contains 1,142,707 works in 13,293 institution–year–degree cells. Repository Tables DEP1-E01–E03 retain accounting.

The broad all-topic doctoral frame includes professional doctorates. The current degree-specific institutional analyses instead use the adopted research-doctorate and master's credential mapping: 663,430 all-topic research-doctorate works and 327,528 master's works before analysis-specific timing/funding restrictions. The remaining broad-frame dispositions are 135,428 other-credential, 229 unresolved and 16,092 unmapped-pair works. Research doctorate denotes PhD or a documented equivalent under the adopted mapping. S3 retains the exact numerator/denominator restrictions and the one timing hold applied to both sides of the institutional comparison.

All 21,073 topical-roster identifiers occurred in the raw all-topic exports. The adopted mapping links 13,219 in-scope works to the retained comparison; eight other-credential works remain outside it but within the scientific population. Matched institutional analyses use compatible institution, completion-year and credential assignments on both sides. Frame membership, outcome eligibility and funding/model availability are separate restrictions (S3; repository support tables).

An exploratory screen compared in-scope master's works with the 2010–2023 all-topic ProQuest frame, using exact normalized full single-author names, the same institutional identifier and a strictly later doctoral completion year. Of 3,460 master's works, 3,422 had usable author names; the denominator retains all 3,460 works. The screen identified 372 candidate pairs representing 371 master's works (10.7%), including 252 of 927 psychology master's works (27.2%). For 214 master's works, a candidate doctoral work was also in the in-scope corpus. The screen included doctoral credentials broadly, rather than restricting candidates to research doctorates. These are candidate identity links, not verified educational histories. A later doctoral work does not establish an en-route master's degree, and no

match does not establish a terminal qualification. Name variation, coincident names, institutional moves and the observation window limit interpretation. S3 reports the corresponding descriptive exclusion sensitivity.

The IPEDS comparison describes indexing independently of topical coding: 100 × summed indexed works / summed awards at matched institutions. Master's theses use all master's awards; flagged research-doctorate dissertations use independently defined research/scholarship doctoral awards. Approved matches and positive available awards are required, retaining awards when no work is indexed and counting each IPEDS identifier once. July–June reporting year is aligned to completion year without a shift. Repository Tables DEP1-E04–E05 and the indexing figure retain annual/pooled results and sensitivities; data notes specify source cells.

The stable doctoral group comprises 411 institutions with positive awards and unchanged constituent IPEDS identifiers in all 14 years; it was not selected on indexing ratios. Its comparison addresses changing institution membership but remains disproportionately NIH-supported. Master's awards include nonthesis degrees, and deposit practices differ. Doctoral classifications, timing and institutional boundaries also differ; ratios above 100 and unexplained extreme comparisons show why these ratios are not capture percentages. The observed indexing trends do not establish the size or direction of bias in topical research. No adjustment factors or funding covariates were derived from them, and the broader doctoral comparison was withheld because missing award rows could not reliably be separated from true zeros.

S1.4 Human reference and measurement performance

Reference construction

The inclusion reference is a fixed sample of 200 records selected without replacement by a prespecified hash ranking from 21,045 assessable candidates. There was no stratification, enrichment, replacement or redraw. Its inclusion probability is 200/21,045; sample retention did not depend on later human judgments or model inclusion. A random 50 of those records received the full six-field unit diagnostic. Disagreement-selected supplementary reviews did not enlarge that probability sample or replace original answers in initial agreement.

The two authors were reference coders, with the main author coordinating (Liss). The main author also developed the codebook and prompts. Both used the same original titles/abstracts and frozen rules. The protocol excluded model outputs, funding groups and historical labels from reading and adjudication packets, and prohibited model assistance or external evidence. The authors attest that they separately returned initial forms before discussion and had not seen the model outputs or the other coder's answers before those forms were frozen. Historical distribution dates are not fully documented.

After frozen initial readings, the authors jointly resolved differences, preserving original entries, evidence, rules, and uncertainty. Independent coding followed by joint resolution has a scientific-abstract precedent (Domínguez-Diaz et al., 2025), which cannot establish this study's independence or correctness. Tier 1 froze on 6 October: 112 in, 84 out and four undetermined. The reference is not error-free truth.

Tier 2 selected the first 100 of the 112 human in-scope records under a separate fixed hash ranking, with no model-status gate or top-up. Its conditional selection probability is 25/28 and joint probability is (200/21,045) × (25/28). Both readers independently supplied C1–C9 for the same frozen consensus scope targets: C1 compares in-scope with other described research, while C2–C9 concern the in-scope target. Initial content agreement is therefore conditional on common targets, not independent end-to-

end human scope/content reliability. Jointly resolved Tier 2 was frozen on 7 October, before model-reference comparison, with 900 decisions: 713 coded, 142 not applicable and 45 cannot determine.

Table S1.4 Achieved reference bases

Reference component	Achieved count or probability	Interpretation
Assessable candidate frame	21,045	Excludes 28 not-assessed abstract exceptions
Tier 1 fixed sample	200	Inclusion probability 200/21,045
Tier 1 consensus scope	112 in; 84 out; 4 undetermined	Reference uncertainty retained
Random detailed unit subset	50 records	Conditional selection probability 50/200
Tier 2 content sample	100 of 112 in-scope; q=25/28	Selection independent of model inclusion
Tier 2 consensus field decisions	900	713 coded; 142 not applicable; 45 cannot determine
Baseline funding groups in Tier 1	140 supported; 60 confirmed zero; 0 unresolved	Group totals, not accuracy denominators

Initial human agreement and uncertainty

Initial agreement uses original separately frozen A/B readings: exact agreement, nominal kappa and Jaccard for multiselects (Artstein and Poesio, 2008). Kappa uses category marginals and may be undefined for constant categories. Jaccard is intersection/union; uncertain, inapplicable and missing responses are not categories or empty sets. Unit alignment used question, topic, purpose and evidence, not identifiers or matching labels. The random 50 yielded 49 one-to-one pairs on 48 records; split/merge groups stayed in segmentation accounting. Raw blanks remain distinct from mechanical rule fills. Repository Tables DEP1-E06–E07 report these human diagnostics and the visible-subset comparison, not model-unit accuracy or an attention correction.

Focus distinguishes Core, Shared and Peripheral/absent; unknown is counted separately. Domain distinguishes drug-only, alcohol-only, both and general/unspecified; the stored label unresolved means the last substantive category, not scope uncertainty or failure. Model-reference focus agreement was 184/199 (92.5%; Wilson 95% CI 87.9–95.4; nominal kappa 0.877, bootstrap 95% CI 0.813–0.934); domain agreement was 109/112 (97.3%; 92.4–99.1; kappa 0.963, 0.915–1.000). Each pipeline retains its own in-scope targets; aggregate agreement does not establish target equivalence. Repository Tables DEP1-E08–E09 retain all endpoints.

Table S1.5 Initial Tier 1 record agreement

Field and comparison base	Exact agreement	Wilson 95% interval	Nominal kappa and bootstrap 95% interval
Scope all 200 three-state readings	191/200; 95.5%	91.7%–97.6%	0.912; 0.853–0.961
Focus jointly coded	185/199; 93.0%	88.5%–95.8%	0.884; 0.823–0.940
Domain jointly in scope	109/110; 99.1%	95.0%–99.8%	0.987; 0.959–1.000

The scope row uses all 200 three-state judgments. Binary agreement among jointly definite human judgments was 190/193, excluding seven uncertainty pairs; those exclusions must not be transferred to the three-state denominator. Focus and domain use their own stated bases. No prevalence-driven category merging or numerical adoption/equivalence threshold was applied.

Table S1.6 Initial Tier 2 content agreement on common human targets

Field	Jointly coded	Exact category or set agreement	Kappa or mean Jaccard and 95% interval	Status agreement
C1 Mixed purpose	99	94/99	Kappa 0.773; 0.608–0.933	100/100
C2 Dissertation type	100	96/100	Kappa 0.875; 0.734–0.973	100/100
C3 Study design	81	78/81	Kappa 0.951; 0.890–1.000	97/100
C4 Level of analysis	94	91/94	Kappa 0.924; 0.832–1.000	99/100
C5 Research objective	100	82/100	Kappa 0.785; 0.690–0.869	100/100
C6 Intervention continuum	20	19/20	Kappa 0.890; 0.589–1.000	95/100
C7 Explanatory domain	48	40/48	Kappa 0.779; 0.626–0.909	91/100
C8 Data type	74	63/74	Mean Jaccard 0.932; 0.892–0.967	91/100
C9 Data source origin	80	78/80	Kappa 0.962; 0.900–1.000	91/100

For Table S1.6, the bootstrap interval accompanies kappa or mean Jaccard, not the exact-agreement fraction. C8 exact agreement compares unordered nonempty label sets. Status agreement is separate from agreement on substantive coded values. These initial A/B endpoints differ from agreement with the adjudicated reference below.

Binary proportions use pointwise 95% Wilson intervals, including zero-error results (Wilson, 1927; Brown et al., 2001). Kappa and mean Jaccard use 5,000 whole-record bootstrap replicates with shared draws across fields and linear percentiles, retaining dependent units. Undefined statistics are not zero; intervals require a defined point and at least 95% defined replicates. No finite-population correction is applied. Tier 2 intervals condition on realized Tier 1 inclusion and Tier 2 selection, omit adjudication uncertainty and provide no simultaneous coverage or guaranteed half-width. Data notes specify implementation and reproduction limits.

Inclusion against the adjudicated reference

The substance-related-criterion comparison retains all 200 reference records and their original production outputs, including model uncertainty and any failure state. Four reference-undetermined cases are excluded from binary accuracy rather than recoded negative. The matrix retains a failure column even though all 200 responses in this sample are validated scientific outcomes.

Table S1.7 Model-reference substance-related criterion matrix

Human reference	Model in	Model out	Model undetermined	Missing, invalid or failed	Total
In scope	112	0	0	0	112
Out of scope	6	77	1	0	84
Undetermined	2	2	0	0	4
Total	120	79	1	0	200

Table S1.8 Substance-related criterion operating characteristics

Measure	n/N	Estimate % (Wilson 95% CI)
Precision on definite reference	112/118	94.9 (89.3–97.6)
Sensitivity	112/112	100.0 (96.7–100.0)
Operational gate specificity	78/84	92.9 (85.3–96.7)
Definite-out specificity	77/84	91.7 (83.8–95.9)
False-inclusion fraction	6/118	5.1 (2.4–10.7)
Missed-inclusion fraction	0/112	0.0 (0.0–3.3)
Three-state exact agreement	189/200	94.5 (90.4–96.9)
Three-state exact agreement, valid model responses	189/200	94.5 (90.4–96.9)
Valid scope-response fraction	200/200	100.0 (98.1–100.0)

Precision uses model inclusions with a definite reference judgment. Sensitivity uses all reference-in records and would count model uncertainty or failure as missed inclusion. Operational-gate specificity is the prespecified non-inclusion measure: it counts 78/84 reference-out records that did not enter the model's included set. Definite-out specificity credits only the 77 explicit model exclusions; the remaining reference-out record was model-undetermined. No missing, invalid or failed components contributed to either estimate. The two specificity measures therefore answer different questions, rather than reporting competing versions of one denominator.

No observed misses is compatible with up to 3.3% missed inclusions in the retrieved frame, from sensitivity's 96.7% lower Wilson limit. Assigning the two uncertain-reference/model-in cases alternatively as incorrect or correct gives precision 112/120–114/120 (93.3–95.0%) and complementary false inclusion 5.0–6.7%. These are assignment bounds, not confidence intervals. No fixed error percentage was subtracted from the population; none estimates search recall.

Research content and usable delivery

All 100 selected human in-scope records were included by the original scope model and have valid original content outputs. There is no received-only subset or offline replacement in this evaluation. The model retains its original choices of in-scope targets, so these are pipeline results rather than isolated classifier accuracy on guaranteed common semantic targets. Initial human content agreement used common human targets and is a different endpoint.

Conditional category agreement uses jointly coded pairs. End-to-end correct delivery uses every selected record with a reference-coded value: model exclusion, uncertainty, failure, missing content, wrong category or wrong applicability would count as a nonmatch. Equal realized selection weights leave these ratios unchanged from unweighted proportions.

Table S1.9 Model content agreement and end-to-end correct delivery

Field	Conditional exact n/N; % (Wilson 95% CI)	End-to-end exact n/N; % (Wilson 95% CI)	Kappa or mean Jaccard (bootstrap 95% CI)
C1 Mixed purpose	88/99; 88.9 (81.2–93.7)	88/99; 88.9 (81.2–93.7)	Kappa 0.442 (0.249–0.625)
C2 Dissertation type	94/100; 94.0 (87.5–97.2)	94/100; 94.0 (87.5–97.2)	Kappa 0.786 (0.605–0.931)
C3 Study design	70/81; 86.4 (77.3–92.2)	70/82; 85.4 (76.1–91.4)	Kappa 0.823 (0.722–0.910)
C4 Level of analysis	84/95; 88.4 (80.4–93.4)	84/95; 88.4 (80.4–93.4)	Kappa 0.733 (0.564–0.868)
C5 Research objective	77/100; 77.0 (67.8–84.2)	77/100; 77.0 (67.8–84.2)	Kappa 0.720 (0.614–0.816)
C6 Intervention continuum	16/20; 80.0 (58.4–91.9)	16/21; 76.2 (54.9–89.4)	Kappa 0.628 (0.294–0.907)
C7 Explanatory domain	32/53; 60.4 (46.9–72.4)	32/53; 60.4 (46.9–72.4)	Kappa 0.468 (0.291–0.634)
C8 Data type	61/78; 78.2 (67.8–85.9)	61/79; 77.2 (66.8–85.1)	Mean Jaccard 0.878 (0.818–0.930)
C9 Data source origin	71/84; 84.5 (75.3–90.7)	71/84; 84.5 (75.3–90.7)	Kappa 0.744 (0.614–0.862)

For C8, exact correctness requires an identical unordered label set; mean Jaccard credits partial overlap and is a different estimand. The conditional/end-to-end differences here arise from applicability, not scope exclusions or failed calls. For study design, 11 category disagreements and one model not-applicable judgment account for 12 nonmatches on 82 reference-coded values. The shared 81-record initial-human and conditional-model comparison base does not by itself establish a formal performance difference between their endpoints.

Field-specific evidence precludes a pooled accuracy claim or formal ranking. Explanatory-domain agreement is 32/53 and intervention-continuum correct delivery 16/21, with wide limits. Mixed-purpose agreement 88/99 coexists with kappa 0.442 because entirely in-scope research dominates: the post hoc diagnostic matches 86/87 such records but only 2/12 mixed-purpose records. This diagnostic does not change codes or primary endpoints. Repository Tables DEP1-E10–E14 retain observability, applicability, all twelve C8 labels and order diagnostics; unobserved labels do not establish detection ability.

Valid delivery does not establish correctness. The model retained reference nonapplicability in only 5/10 study-design, 0/5 level-of-analysis and 4/9 data-type cells. Its definite answers cannot resolve 45 uncertain reference cells; no field-level cannot-determine response was supplied. Confidence summaries are descriptive, not calibrated correctness probabilities. Uncertainty, low confidence and candidate explanations were not used to tune codes, exclude cases or select a better rerun. Repository Tables DEP1-E16–E17 distinguish overlapping discrepancy events from independent errors.

Accuracy is not established uniformly across degree, field, institution, period or funding groups. The funding diagnostic uses fixed FY2005–2009 institutional support, not S3 rolling windows: precision was

76/82 (92.7%; Wilson 95% CI 84.9–96.6) in baseline-supported cases and 36/36 (100%; 90.4–100) in confirmed-zero cases (repository Table DEP1-E15). These estimates establish neither equality nor differential error under later windows. Downstream intervals omit classification, retrieval and linkage uncertainty. Failed development repeatability is a separate limitation. Computational verification supports calculations, not reference infallibility, historical blinding or independent award-link validation.

S1.5 Award-linked search capture

The award-linked check used eight existing NIDA/NIAAA RePORTER files for FY2005–2023 and the all-topic U.S. English-language dissertation frame for 2010–2023. Repeated fiscal-year observations were reconciled into 1,722 award cores, 1,720 source-person clusters and 1,724 award/person relationships. F30/F31 and R36 identify student PD/PIs under their mechanism-specific roster rules. The dissertation author was the target; an advisor-only name match was insufficient. No additional NIH pull was used.

Name blocking used normalized surnames and first initials. Clear links required full given-name agreement or a source-verified alias, compatible institution/timing and resolution of plausible competitors using recipient histories and reviewed institutional mappings. Middle-name/suffix conflicts, historical institution changes and competing works were reviewed. Phonetic agreement or a later affiliation alone was insufficient. The primary completion window extended from the first observed funded fiscal year through the last observed funded year plus three, inclusive; plus-two and plus-five years were sensitivities. Source holds remained explicit.

Repository Table DEP1-E18 preserves historical award/person accounting. No match means no identified link under these rules, not certified absence of a dissertation or NIH support. 820 primary matched works, all present in the all-topic ProQuest frame. The base comprises 786 doctoral and 34 master's works, with 582 and 19 included. These broad frame levels differ from S3's research-doctorate subset. Matching a work to an award holder does not establish that the award funded that degree. Works are deduplicated across repeated awards and mechanisms; F30/F31 and R36 overlap by two works.

Search capture is entry into the topical candidate frame before abstract-assessability or coding decisions. It was 598/775 (77.2%) for F30/F31 and 39/47 (83.0%) for R36. Final inclusion was 567/775 (73.2%; Wilson 95% CI 69.9%–76.2%) and 34/47 (72.3%; 58.2%–83.1%), respectively. The pooled union is secondary accounting: 636/820 captured and 601/820 included. Table S1.10 retains both clear primary links and the clear-plus-possible sensitivity; those definitions are not bounds on true coverage.

Table S1.10 Final inclusion and pipeline dispositions in the primary timing window

A Clear links

Mechanism	Matched works	Included	Not retrieved	Retrieved excluded	No usable abstract	Included %
F30	80	59	18	1	2	73.8
F31	695	508	159	27	1	73.1
F30/F31 union	775	567	177	28	3	73.2
R36	47	34	8	5	0	72.3
All mechanisms, union	820	601	184	32	3	73.3

B Clear plus possible links

Mechanism	Matched works	Included	Not retrieved	Retrieved excluded	No usable abstract	Included %
F30	84	62	19	1	2	73.8
F31	737	540	168	28	1	73.3
F30/F31 union	821	602	187	29	3	73.3
R36	49	35	9	5	0	71.4
All mechanisms, union	868	637	195	33	3	73.4

Within each row, included, not retrieved, retrieved-but-excluded and no-usable-abstract counts partition the matched works. Scientific-undetermined and incomplete-processing counts are zero here and remain separately defined in the supporting exports. Of 636 retrieved works, 633 have assessable abstracts. The 184 nonretrieved works account for 84.0% of the 219 pooled works not included. Nonretrieval means absence from the topical query despite presence in the all-topic frame; their substantive scope is unknown. The 32 retrieved exclusions are adopted scope decisions, not independently established errors. A current criterion-level exclusion breakdown is not inferred from these disposition totals.

Repository Tables DEP1-E19–E20 retain timing and period comparisons. Each timing scenario recomputes the candidate graph; wider windows can add competitors and reduce clear matches. Period-dependent capture may affect research profiles, while award mix, linked-work composition and window edges also vary.

S2 Research content

S2.1 Population outcomes and denominators

We compared research content across training settings in 13,227 distinct in-scope works completed in 2010–2023: 5,816 in 2010–2016 and 7,411 in 2017–2023. Table S2.1 describes the full corpus before content eligibility or model-support restrictions.

Table S2.1 Degree and field composition of the full corpus

Broad field	Master's n = 3,460	Research doctorate n = 7,964	PsyD n = 807	DrPH n = 186	Other named doctorate n = 781	Other or unresolved credential n = 29	All works n (%)
Psychology	7.0	14.8	4.8	0.0	<0.1	<0.1	3,535 (26.7)
Public / population health	2.9	6.6	0.0	1.3	0.4	0.0	1,483 (11.2)
Clinical health	1.1	2.7	0.0	<0.1	2.5	0.0	838 (6.3)
Biomedical / life sciences	3.7	11.9	0.0	0.0	<0.1	<0.1	2,068 (15.6)
Chemical / physical sciences	0.8	1.7	0.0	0.0	0.0	0.0	329 (2.5)
Social work / counseling	1.5	3.2	<0.1	0.0	0.2	<0.1	658 (5.0)
Social sciences / policy	3.5	7.1	0.4	<0.1	0.7	<0.1	1,540 (11.6)
Education	0.3	1.3	<0.1	0.0	0.9	<0.1	334 (2.5)
Engineering / computing	0.7	1.1	0.0	0.0	<0.1	0.0	232 (1.8)
Humanities / arts	0.7	0.6	0.0	0.0	0.2	0.0	206 (1.6)
Interdisciplinary / other	0.3	0.5	0.1	0.0	<0.1	0.0	134 (1.0)
Field unavailable	3.8	8.7	0.7	<0.1	0.8	<0.1	1,870 (14.1)
All fields	26.2	60.2	6.1	1.4	5.9	0.2	13,227 (100.0)

The inclusion definition covers substance-related and endogenous-system research, rather than the full mission of either NIDA or NIAAA. General/unspecified substance is a heterogeneous residual category that includes unresolved substance-class attribution; it does not identify a uniform group of general addiction studies. Addiction/substance-use focus was classified separately as core, shared or peripheral/absent and did not determine inclusion. Institutional analyses were not restricted to core-focus works because a consistent focus classification was unavailable for their expanded population.

Credential columns refer to each work's adopted degree classification. Research doctorate comprises PhD and documented equivalents; other named doctorates retain determinate credentials other than PsyD and DrPH. Other/unresolved credential retains ambiguous or mixed assignments and specialist/postmaster's or cross-level credentials. Field unavailable is separate from interdisciplinary/other.

Cells and All fields percentages use all 13,227 works, not column totals; headers give credential totals and the last column field counts/shares. Rounding may prevent exact margin sums. <0.1 denotes a positive share below 0.1%; 0.0 is zero. Repository Table DEP2 T20 retains exact counts and shares.

The four outcomes are study design (C3; 12 categories), research objective (C5; 12), level of analysis (C4; eight), and data type (C8; 12 nonexclusive indicators), under the revision-15 definitions and S1 coding procedures. C3, C5 and C4 are mutually exclusive within their applicable, determinate populations. Missing, undetermined, inapplicable and operational states are separate from substantive categories. Chemical synthesis or characterization is a design category, not literature synthesis. C8 records up to three central forms of data actually analyzed: valid unselected indicators are zero, whereas an ineligible or missing list is not twelve zeroes. Its percentages overlap and do not sum to 100%. Data form is neither data provenance nor a complete inventory of methods or skills; the absence of a capacity flag does not establish that no additional forms were omitted.

Table S2.2 Outcome-specific distinct-work populations

Outcome	Determinate	Inapplicable	Undetermined	Group-model eligible	Fixed top-50 eligible
C3	12,538	672	17	10,645	5,824
C4	13,225	0	2	11,198	6,105
C5	13,226	0	1	11,198	6,105
C8	12,947	262	18	10,972	5,998

Group models require known adopted degree, field and institution; fixed-contributor institutional models require outcome eligibility independently of group-predictor eligibility (Table S2.2; repository Tables DEP2 T1–T2, T12–T14). The 14 October content fields remain frozen; later award-link adoption changed none. The optional priority comparison is outside this distribution and uses all in-scope works, including unknown/inapplicable codes, so its denominator differs.

S2.2 Group profiles and comparisons

Master's credentials are pooled within broad field; research doctorates combine PhD and documented equivalents, with PsyD and DrPH distinct. Interdisciplinary/other fields remain in coverage accounting but outside group inference. Nonexistent combinations are not predicted; unsupported observed combinations remain in repository coverage tables.

Mean-bias-reduced multinomial models describe observed degree–field/period cells for C3, C5 and C4; twelve marginal binomial models describe C8. For category count y, eligible n and K exclusive categories, the group-cell probability is (y+0.5)/(n+K/2); each C8 prevalence is (y+0.5)/(n+1). Within-period works receive empirical weights; combined profiles average earlier/later probabilities 0.5/0.5, independently of cohort size. Analyses used R 4.6.1 and brglm2 1.1.1; data notes specify numerical implementation.

Reporting requires at least 50 eligible works and ten contributing institutions in each period, separately by outcome. The same 16 groups qualify for all outcomes: master's, research doctorate and PsyD in psychology; master's, research doctorate and DrPH in public/population health; master's and research doctorate in clinical health, biomedical/life sciences, social work/counseling and social sciences/policy; and research doctorates in chemical/physical sciences and education. Support membership remains fixed during resampling.

Every unordered pair occurs once: 120 per outcome, 5,280 category contrasts, including the 264 inherited comparisons. Contrasts are 100 × (group A minus group B) in frozen order, including master's-minus-research-doctorate directions. Cross-field/cross-degree pairs describe training settings jointly and cannot isolate degree effects or rank degree against field. Repository Tables DEP2 T3–T7 retain profiles, bases, directions, intervals and unavailable states; Figure DEP2 F1 shows all supported groups, including PsyD and DrPH.

For example, master's nonrandomized-experiment profiles were 26.2% in psychology and 77.4% in biomedical/life sciences: −51.2 percentage points (95% pointwise interval −65.5 to −36.6). Within psychology, intervention objectives were 7.3% for master's and 15.5% for research doctorates: −8.2 points (−11.4 to −4.6); an intervention objective need not imply experimental design. Biomedical/life-sciences biological/physiological-data profiles were 56.9% and 74.5%: −17.6 points (−30.4 to −3.6). Data forms overlap. Small differences or intervals spanning zero do not establish equivalence; repository tables retain every comparison.

S2.3 Temporal comparisons

Primary change is 100 times the later-minus-earlier fitted probability. There are 704 supported category/form changes. The continuing-institution sensitivity retains contributors with at least one eligible work in both periods within each outcome and group, then reapplies the 50-work and ten-institution floor. It supplies 560 supported changes and 144 unsupported counterparts among those 704 targets. Full registries also retain unsupported combinations and reasons.

Social sciences/policy community/jurisdiction/population conclusions increased from 9.5% to 20.7% among master's works (+11.2 points; 95% pointwise interval 5.1–17.5), versus +7.4 among continuing institutions. Research-doctorate profiles rose 11.4% to 24.8% (+13.4; 8.6–18.0), also +13.4 at continuing institutions. Master's change attenuates; doctoral change is similar. Institutional shares, eligibility and within-institution mixtures can still change. These are different work cohorts, not longitudinal trainees. No period-specific pairwise family or between-group change contrast was added. Repository Tables DEP2 T8–T9 and Figures F3–F6 retain small, reversed, unsupported and unavailable changes.

S2.4 Institutional portfolios and held-out gain

The 50 largest full-window contributors were fixed before outcome exclusions, with ties resolved by stable institution identifier. Their canonical definitions and membership remain fixed; 6,105 works yield Table S2.2's eligible populations. Institutional models use additive institution/period mean-bias-reduced multinomial models for C3/C5/C4 and marginal binomial models for C8, with a common log-odds period effect and equal-period averaging. They retain degree/field composition and estimate neither

composition-adjusted nor causal institutional contributions. Repository Figure DEP2 F2 and Tables T10, T12 and T14 retain every contributor's counts and unavailable intervals.

Held-out distinctiveness covers C3/C5/C4 only. Institution-plus-period and pooled period-only models use identical five-fold splits repeated five times within institution-period strata: sorted work identifiers are deterministically shuffled and assigned round-robin. Institutions require at least five works per period and training representation; all 50 qualify. Both models use the same exercise-eligible population. Each work is evaluated once per repeat; repeats are not independent samples.

Gain is the log probability of a work's observed category under the institution-plus-period model minus that under the pooled model, in nats per work. Period means receive equal weights within institution, followed by equal weighting of institutions and repeats. The work-weighted companion pools held-out work gains within repeat before averaging repeats. We do not score averaged predictions or treat folds as independent observations.

Table S2.3 Retained held-out exercise populations and observed gains

Outcome	Earlier n	Later n	Equal-institution gain	Work-weighted gain
C3	2,594	3,230	0.160	0.205
C5	2,711	3,394	0.050	0.068
C4	2,711	3,394	0.107	0.120

These descriptive points have no inferential intervals. Some institutional gains are negative; aggregate gain establishes neither universal benefit, future performance nor rankings. Different category systems, baselines and populations preclude outcome rankings. C8 has no joint institutional prediction target. Repository Tables DEP2 T11 and T17–T19 retain actual split assignments and training/evaluation counts, rather than assumed equal fold sizes.

S2.5 Uncertainty and interpretation

Group intervals use institution-cluster bootstrap draws, retaining each sampled institution's eligible works across periods and groups with cluster multiplicity. Primary and continuing populations have separate deterministic streams. Interpretation requires comparable institution clusters under the indexing and coding process and does not establish complete ProQuest coverage. Institutional-profile intervals instead resample works within fixed institution-period strata, conditional on stratum sizes and selected institutions.

Group runs attempted 10,000 draws per outcome; institutional profiles attempted 5,000. Pairwise endpoints share replicate identity; missing endpoints make contrasts unavailable. Marginal limits are not subtracted, unrelated draws not paired, and failures not replaced. All 5,280 contrasts have 10,000 usable attempts. Type-7 2.5th/97.5th percentiles require at least 95% successful attempts. Endpoint shifts between full streams and 8,000/4,000-attempt prefixes must be ≤0.005 in probability or 0.5 percentage points. These Monte Carlo checks do not validate interval coverage. Master seed: 20261014.

Reportable bounds are unavailable for 63 supported nonprediction targets: 60 institutional-profile cells, one earlier C8 point, one later C4 point and one C3 change. Repository results retain their points/reasons; diagnostic bounds do not replace them. The 156 held-out scores have no intervals: historical bootstrap checks found predictive-score centering/coverage problems. Numerical stability does not resolve them; no replacement interval study is claimed.

Intervals are conditional and pointwise. They condition on coded and linked data, provide no simultaneous coverage across 5,280 contrasts, and omit uncertainty from retrieval, coding, adjudication, linkage, field mapping and fixed-top-50 selection. Section S1.4 discusses differential classification error and distinguishes accuracy from repeatability. We use no omnibus tests, multiplicity-adjusted discovery claims, significance stars or selection solely by exclusion of zero.

Support does not ensure diffuse institutional contribution or adequate power. Contributor deletions are influence diagnostics, not new confidence intervals. The previously examined C4 organization/program degree gap in social work/counseling changes sign under some deletions and is not used as a broad degree claim. Fifteen inherited C3-coded works have C8 inapplicable; four empirical/no-empirical-data flags and six data-type/unknown-source flags remain. These are unresolved adjudication cases, not established errors or authorization to recode the frozen snapshot. C3/C5 agreement does not validate C4/C8 accuracy, and this assembly adds no human-reference evaluation.

Results concern indexed, retrieved works under adopted eligibility/mapping rules, not acquired skills, training quality, causal program effects or unique graduates. Selecting an institution does not select a joint degree/field population; category selection retains the eligible-outcome denominator. S1 states search and measurement limitations. The repository catalog supplies Tables DEP2 T1–T20 and six figures with captions; reproduction instructions define the aggregate point-rebuild script's scope.

S3 Funding connections and opioid research

Institutional output, distinct-work funding connections, same-investigator participation and exploratory opioid comparisons have different units, exposures and populations. They cannot provide interchangeable funding-effect estimates. The repository catalog retains complete DEP3 results, figures and diagnostics; data notes and reproduction instructions specify each family's calculation and replay scope.

S3.1 Institutional funding and research output

Cohorts, exposures and units

The adopted population contains 13,227 distinct scientific works. The degree comparisons retain 7,964 research-doctorate and 3,460 master's works; 1,803 other-credential works remain in overall accounting. Research doctorate means PhD or a documented research-doctorate equivalent under the adopted credential mapping. All eight works outside the institutional frame have other credentials. Degree labels describe the credential of a work, not an author's highest qualification. Tables S3.1–S3.4 separate the cohort, design units, connection accounting and outcome-eligible bases.

Table S3.1 Current adopted work cohorts

Credential group	Full works	Historical for-profit exclusion	Outside institution frame
Research doctorate	7,964	7,389	0
Master's	3,460	3,391	0
Excluded other credential	1,803	1,363	8

Table S3.2 Analysis units

Design	Unit and denominator
Institution	Institution-year-degree counts; exact mapped-degree all-topic output D.
D: combined work connection	Distinct scientific works; each counted once. Primary: preceding-funded advisor/committee or clear individual award, or both. Comparator: all remaining outcome-eligible works.
B: within PI	PI-institution-year risk-set rows and distinct in-scope PI-work participations. One work can contribute to multiple PIs; it is not a unique-student count.

Table S3.3 Observed role-field and ownership coverage

Degree	Observed works	Advisor / committee component	Combined primary	Incomplete roles	Unknown ownership
Doctorate	7,964	3,144	3,236	1,349	34
Master's	3,460	755	758	782	3

Table S3.4 Outcome coverage and descriptive support

Family	Degree	Combined connected coded works	Combined remaining coded works	Institution timing-resolved coded works
C3	Doctorate	3,214	4,492	7,705
C3	Master's	751	2,398	3,149
C4	Doctorate	3,235	4,728	7,962
C4	Master's	758	2,701	3,459
C5	Doctorate	3,235	4,728	7,962
C5	Master's	758	2,702	3,460
C8	Doctorate	3,230	4,639	7,868
C8	Master's	756	2,552	3,308

Institutional all-topic denominators use the same credential mapping as the in-scope numerators: 663,430 research-doctorate and 327,528 master's works before analysis-specific restrictions. Other, unresolved and unmapped credentials remain explicit exclusions, as described in S1. A single completion-year hold is removed from both timing-dependent numerator and compatible denominator; the work remains in observed distinct-work accounting. Unresolved funding is missing, not zero, and 28 unresolved institutional exposure cells remain outside funding regressions. Outcome eligibility is separate: C3/C4/C5 require an applicable, determinate code; C8 requires a valid multi-response list, in which unselected forms are zero. Missing, inapplicable or undetermined lists are not zero profiles.

Institutional exposure counts distinct funded application IDs within each fiscal year, pooling NIDA/NIAAA and documented recipient or partner routes, then sums those annual counts. Repeated applications contribute application-years; this is neither unique applications over the whole window nor funding dollars. Research-doctorate histories cover the five preceding fiscal years and master's histories the two preceding years, excluding completion fiscal year. Dividing by five or two gives annualized exposure. The common contrast from one to four annualized applications corresponds to five versus twenty application-years for doctorates and two versus eight for master's works. Unequal windows qualify every degree comparison. Documented predecessor and joint-program routes are retained without globally assigning historical funding to a later institutional name.

Institutional control uses contemporaneous IPEDS directories (National Center for Education Statistics, n.d.). Whole institutions ever identified as for-profit are excluded in the ownership sensitivity; unknown ownership stays included, so this is neither verified nonprofit coverage nor independent replication. Data notes retain historical Carnegie edition assignment and source construction.

Count models and inference

For institution i, completion year t and degree d, the PPML model is log $E(S_{itd}|X)=\alpha_{id}+\lambda_{td}+\beta_d \log(1+A_{itd})+\gamma_d \log D_{itd}$, where S is in-scope count, D positive exact mapped-degree all-topic output, and A annualized preceding funding. The output coefficient is estimated, not an offset. Zero outcomes remain until explicit all-zero-effect or separation trimming. With $\Delta=\log(5)-\log(2)$, $RR_d=\exp(\beta_d\Delta)$. The

direct ratio is exp[(β_M,common−β_RD,common)Δ] from iterated common retained institution-year cells with separate degree nuisance parameters, not headline-fit division.

Table S3.5 Adopted count estimates and institution-cluster uncertainty

Ownership	Degree target	RR or ratio of RRs	Pointwise 95% CI	Three-target simultaneous 95% CI
Full	Research doctorate	1.155	1.036 to 1.289	1.010 to 1.321
Full	Master's	1.135	0.920 to 1.400	0.875 to 1.471
Full	Master's / doctorate (direct)	0.976	0.775 to 1.228	0.735 to 1.294
For-profit excluded	Research doctorate	1.153	1.033 to 1.286	1.008 to 1.319
For-profit excluded	Master's	1.135	0.919 to 1.402	0.875 to 1.473
For-profit excluded	Master's / doctorate (direct)	0.976	0.775 to 1.228	0.735 to 1.295

Table S3.6 Main and common-reference retained bases

Ownership	Degree	Cells	Institutions	In-scope S	Mapped output D
Full	Doctorate	4,444	350	7,949	649,632
Full	Master's	3,354	283	3,455	307,177
Full common	Doctorate	2,440	196	4,983	399,094
Full common	Master's	2,440	196	2,759	262,338
Excluded	Doctorate	4,304	337	7,374	625,090
Excluded	Master's	3,309	279	3,386	305,789
Excluded common	Doctorate	2,408	193	4,905	396,803
Excluded common	Master's	2,408	193	2,714	261,076

Doctorate inference is positive; master's and the direct common-population ratio are imprecise. Different significance decisions do not establish a degree difference, nor similar points equivalence. The six count targets use institution-grouped innovations and exact intercept profiling. For profile information H, institution information H_g, score s_g and contrast c, variance is (G−1)/G × Σ_g[s_g′(H−H_g)^−1c]². Working-Poisson degrees of freedom are tr(K)²/tr(K²), retaining nuisance estimation. Approximate t intervals accompany three-target Bonferroni families within each ownership

population. This profile-HC3 reporting policy followed inspection of points and validation; it was not prespecified. Independent institutional innovations and an adequate conditional mean remain assumptions, not exact coverage guarantees.

Only eleven doctorate and nineteen master's institutions bracket both annualized endpoints; six common-reference institutions do so for both degrees. Bracketing describes support, not conditional positivity; the common slope applies where both endpoints were unobserved. Repository Tables DEP3-E02–E04 and count diagnostics retain algebraic translations, deletion ranges, calibration and nonadopted inference; none is independent confirmation or selected by significance.

Table S3.7 Observed support for the annualized 1 to 4 contrast

Degree	Observed support measure	Cells	Institutions
Doctorate	Exactly 1 application ID/FY	101	51
Doctorate	Exactly 4 application IDs/FY	33	24
Doctorate	Unit range brackets 1 to 4	149	11
Master's	Exactly 1 application ID/FY	212	80
Master's	Exactly 4 application IDs/FY	46	26
Master's	Unit range brackets 1 to 4	244	19

The timing check jointly includes preceding and strictly later funding, excluding completion fiscal year from both. Later histories require complete source and ownership evidence through fiscal year 2023; missing histories are not zero. The preceding-only comparison uses exactly the same retained cells as its joint model. Positive later associations and the master's preceding/later point ratio below one leave temporal direction unresolved.

Table S3.8 Joint preceding and later funding on compatible retained cells

Degree	Retained cells	Preceding RR (joint model)	Later RR (joint model)	Preceding-only RR (same cells)
Doctorate	2,575	1.207	1.134	1.167
Master's	2,769	1.097	1.121	1.104

Institutional content models retain all 44 categories/forms, both degrees, compatible direct comparisons and both ownership populations: 528 targets, with 488 available points and 40 unavailable states. Category-count models retain the estimated log all-topic-output coefficient. Portfolio-rate models instead offset log outcome-eligible in-scope output and omit zero eligible portfolios; they do not also estimate log all-topic output. Their selected samples differ, so count and share associations are not an exact decomposition. Portfolio-rate PPML is unbounded: fitted category rates can exceed one, especially under sparse-support endpoint extrapolation. They are conditional relative rate associations, not calibrated probabilities; no clipping or renormalization is applied. All primary institutional-content intervals remain unavailable.

Repository Table DEP3-E03 retains all selective checks: thirty points and six unavailable current-core-focus targets do not inherit count inference. Legacy focus cannot restrict the expanded population. Own/other institute contrasts do not establish specificity. Nonadopted method-transfer intervals retain

eighteen targets per ownership population; no adjusted p value is below .05. Indexed-output associations cannot distinguish capacity from index capture. All apply to deposited examples and calculations.

S3.2 Distinct work reach and research content

A primary connection is a clear preceding-funded advisor or committee member, a clear individual-award association, or both. Each work counts once regardless of the number of people, roles or awards. Advisor/committee histories use five preceding fiscal years for research doctorates and two for master's works, excluding completion year. Individual-award eligibility extends from first observed funding fiscal year through last observed funding fiscal year plus three, inclusive and bounded to 2010–2023; it can include completion-year funding and does not use lifetime project-end dates. The possible-award sensitivity changes only the award component, keeping clear advisor/committee connections fixed.

Remaining works have no identified connection under these rules; they are not verified unfunded works. Missing roles can coexist with a connection through another observed role or an award. Tables S3.9 and S3.10 retain primary, component and sensitivity definitions, with the full population followed by the historical-for-profit exclusion. The all-credential population contains both retained degree subsets. These are adopted-corpus descriptions without inferential intervals.

Table S3.9 Observed distinct-work reach: full population

Population	Total works	Advisor / committee component	Combined primary	Possible-award sensitivity
Research doctorate	7,964	3,144 (39.5)	3,236 (40.6)	3,239 (40.7)
Master's	3,460	755 (21.8)	758 (21.9)	758 (21.9)
Other credentials	1,803	73 (4.0)	73 (4.0)	73 (4.0)
All credentials	13,227	3,972 (30.0)	4,067 (30.7)	4,070 (30.8)

Table S3.10 Observed distinct-work reach: excluding institutions ever identified as for-profit

Population	Total works	Advisor / committee component	Combined primary	Possible-award sensitivity
Research doctorate	7,389	3,143 (42.5)	3,235 (43.8)	3,238 (43.8)
Master's	3,391	755 (22.3)	758 (22.4)	758 (22.4)
Other credentials	1,363	73 (5.4)	73 (5.4)	73 (5.4)
All credentials	12,143	3,971 (32.7)	4,066 (33.5)	4,069 (33.5)

Route categories partition the connected union; institute counts overlap. Advisor/committee connections reach 3,972 works, clear awards 601 and both routes 506: the union is 4,067/13,227 (30.7%). The possible-award sensitivity is separate (30.8%). NIDA reaches 3,161 works, NIAAA 1,836, both institutes 930. Repository overlap and complete-role sensitivity tables retain ownership variants.

Table S3.11 Connection route overlap

Population	Advisor/committee without award	Both routes	Award without advisor/committee	Union
Research doctorate	2,657	487	92	3,236
Master's	739	16	3	758
Other credentials	70	3	0	73
All credentials	3,466	506	95	4,067

Table S3.12 Institute overlap

Population	NIDA	NIAAA	Both institutes	Either institute
Research doctorate	2,605	1,422	791	3,236
Master's	491	394	127	758
Other credentials	65	20	12	73
All credentials	3,161	1,836	930	4,067

Tables S3.13 and S3.14 retain every degree-field reach row, including field unavailable, sparse positive cells and the observed 0/35 master's education cell. Each denominator is all works in that degree-field group. NIDA plus NIAAA minus both equals the combined union. Reach differs from field composition within a connection group and from content percentages among outcome-eligible works. Sparse observed differences do not establish precise field rankings.

Table S3.13 Degree and field funding reach: research doctorates

Field	All works	Combined n (%)	NIDA n (%)	NIAAA n (%)	Both n
Psychology	1,958	877 (44.8)	643 (32.8)	525 (26.8)	291
Public and population health	879	317 (36.1)	265 (30.1)	104 (11.8)	52
Medicine, nursing, and clinical health professions	355	102 (28.7)	83 (23.4)	39 (11.0)	20
Biomedical, life, neuroscience, and pharmaceutical sciences	1,579	1,142 (72.3)	958 (60.7)	479 (30.3)	295
Chemical and physical sciences	225	91 (40.4)	80 (35.6)	14 (6.2)	3
Social work, counseling, and human services	419	92 (22.0)	81 (19.3)	29 (6.9)	18
Social sciences, policy, administration, business, justice, and communication	938	142 (15.1)	121 (12.9)	56 (6.0)	35
Education	169	10 (5.9)	7 (4.1)	3 (1.8)	0
Engineering, computing, and quantitative sciences	141	41 (29.1)	32 (22.7)	18 (12.8)	9
Humanities and arts	84	1 (1.2)	1 (1.2)	0 (0.0)	0
Interdisciplinary or other substantive field	62	24 (38.7)	23 (37.1)	7 (11.3)	6
Field unavailable	1,155	397 (34.4)	311 (26.9)	148 (12.8)	62

Table S3.14 Degree and field funding reach: master's works

Field	All works	Combined n (%)	NIDA n (%)	NIAAA n (%)	Both n
Psychology	927	369 (39.8)	218 (23.5)	228 (24.6)	77
Public and population health	381	85 (22.3)	68 (17.8)	26 (6.8)	9
Medicine, nursing, and clinical health professions	145	17 (11.7)	11 (7.6)	7 (4.8)	1
Biomedical, life, neuroscience, and pharmaceutical sciences	485	166 (34.2)	119 (24.5)	79 (16.3)	32
Chemical and physical sciences	104	8 (7.7)	8 (7.7)	0 (0.0)	0
Social work, counseling, and human services	194	7 (3.6)	4 (2.1)	4 (2.1)	1
Social sciences, policy, administration, business, justice, and communication	459	25 (5.4)	17 (3.7)	10 (2.2)	2
Education	35	0 (0.0)	0 (0.0)	0 (0.0)	0
Engineering, computing, and quantitative sciences	90	15 (16.7)	9 (10.0)	6 (6.7)	0
Humanities and arts	96	1 (1.0)	1 (1.0)	0 (0.0)	0
Interdisciplinary or other substantive field	43	6 (14.0)	4 (9.3)	2 (4.7)	0
Field unavailable	501	59 (11.8)	32 (6.4)	32 (6.4)	5

Different works by the same author remain separate observations. Using the exploratory candidate links described in S1.3, excluding the 252 psychology master's works with a later same-institution doctoral candidate removed 110 works with an identified NIDA/NIAAA connection. The connection share was 39.8% (369/927) before exclusion and 38.4% (259/675) afterward, a decrease of 1.4 percentage points. This descriptive exclusion does not establish equivalence or correct for verified person-level overlap. Candidate links do not establish educational histories, and the degree comparisons concern works rather than disjoint populations of people.

Content contrasts are connected-minus-remaining percentages on outcome-eligible bases; direct degree contrasts subtract doctorate from master's connection differences, retaining all four denominators. Repository Figure DEP3 F2 and tables retain all 616 targets. Highlighted explanatory categories were selected after inspection. Connected works have larger pooled shares of nonrandomized experiments, molecular/cellular and nonhuman-organism research, behavioral-task and biological/physiological data in both degrees; these combine field composition and within-field differences.

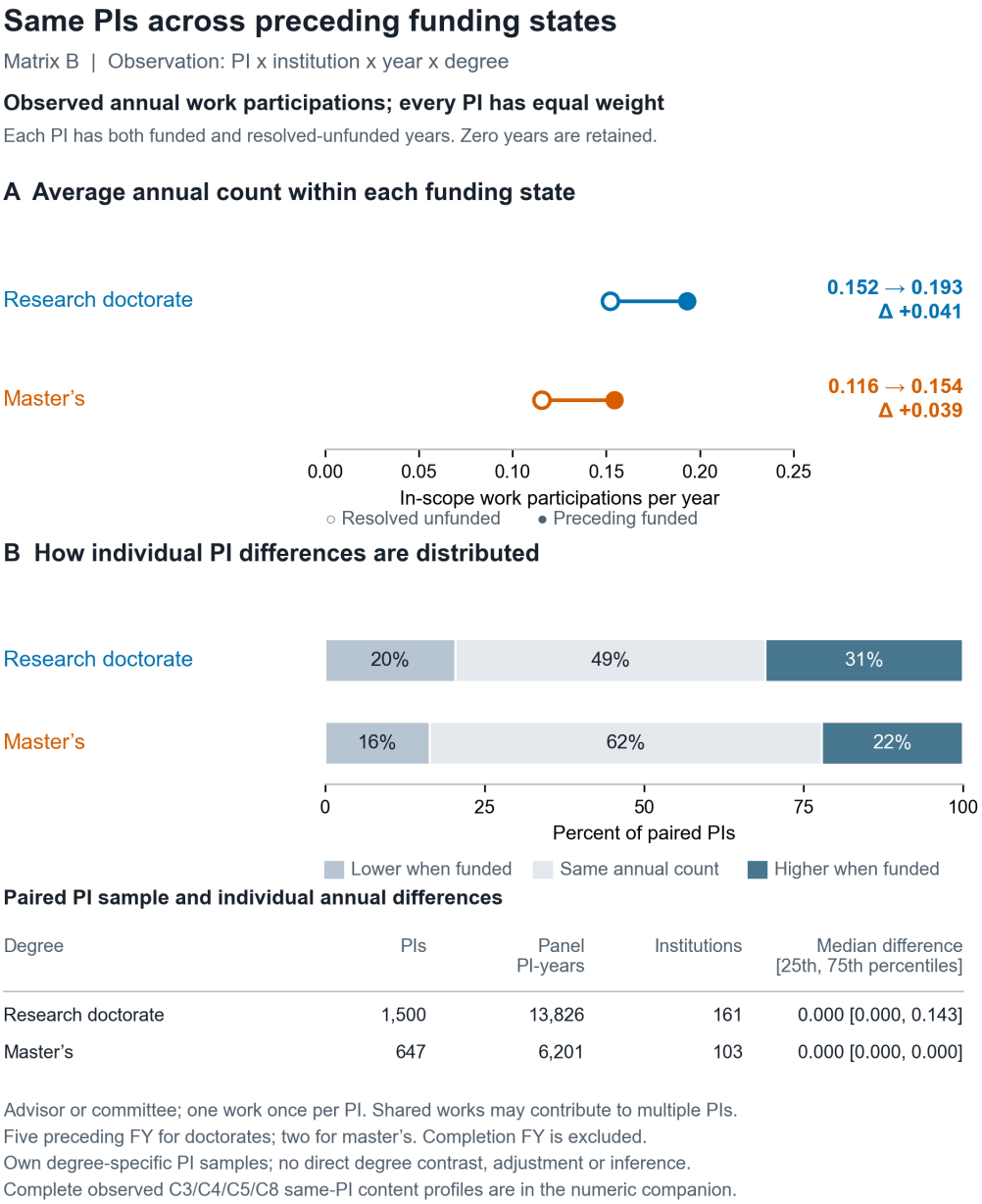
Repository Figure DEP3 F3 compares full crude, common-support crude and common-field summaries. Both groups receive pooled eligible field weights $w_f=N_f/\Sigma_f N_f$ on strata containing both; the share is $\Sigma_f w_f p_{gf}$. All twelve doctorate strata have support. Omitting master's education only from supported comparisons retains 98.95–99.00% of eligible works; field unavailable stays explicit. Reweighting can reduce or reverse pooled contrasts. This descriptive arithmetic has no intervals and controls neither institution, period nor within-field differences; no causal effect is identified. Tables retain counts, weights, omitted strata and sparse flags. The optional priority demonstration is outside this distribution and separate from opioid comparisons.

S3.3 Investigator participation

The same-investigator design counts each work once per clearly linked advisor or committee investigator. A shared work can contribute to several investigators; these are investigator–work participations, not unique students or training slots. A PI/institution risk set spans the first to last clear appearance across retained degrees, requiring that the target degree was observed. Internal eligible years retain zero-participation rows. Exclusions cover source-wide movers, unresolved person/name-years or preceding funding, and insufficient institution-degree-year role coverage. Coverage requires at least 25 mapped-degree works, 95% parsed advisor fields and 80% parsed committee fields. Internal years do not establish uninterrupted residence or zero training opportunity.

The descriptive comparison selects PIs with at least one eligible preceding-funded year and one eligible resolved-unfunded year, including eligible PIs with no in-scope works in either state. For each PI, funded and unfunded participations are divided by that PI's eligible years in the respective state. The paired difference is then averaged with equal PI weights. Different state-year totals affect denominators, not PI weights. The degree groups use separate PI populations, with 1,500 doctorate PIs contributing 13,826 eligible PI-years and 647 master's PIs contributing 6,201. Both median paired differences are zero. The table and figure retain means, state support and the distribution of individual differences.

Figure S3.1 Same-PI participation across funding states



Paired PI sample and individual annual differences

Degree	PIs	Panel PI-years	Institutions	Median difference [25th, 75th percentiles]
Research doctorate	1,500	13,826	161	0.000 [0.000, 0.143]
Master's	647	6,201	103	0.000 [0.000, 0.000]

Advisor or committee; one work once per PI. Shared works may contribute to multiple PIs.
Five preceding FY for doctorates; two for master's. Completion FY is excluded.
Own degree-specific PI samples; no direct degree contrast, adjustment or inference.
Complete observed C3/C4/C5/C8 same-PI content profiles are in the numeric companion.

Each eligible PI supplies annual participations in preceding-funded and resolved-unfunded states. Panel A averages state rates and paired differences equally across PIs; B shows individual differences. Eligible zero years and all-zero PIs remain. Degree samples differ; works can contribute to several PIs. Full and ownership-excluded inputs are identical. Doctorate/master's windows cover five/two preceding fiscal years, excluding completion year. No modeled adjustment, intervals or causal interpretation is applied. Repository investigator tables retain support and content profiles.

Table S3.15 Descriptive same-PI annual participation

Ownership	Degree	PIs	Funded mean per year	Unfunded mean per year	Mean paired difference
Full	Research doctorate	1,500	0.193	0.152	0.041
Full	Master's	647	0.154	0.116	0.039
For-profit excluded	Research doctorate	1,500	0.193	0.152	0.041
For-profit excluded	Master's	647	0.154	0.116	0.039

Same-PI content requires positive eligible participations in both states. Absent categories are zero; no eligible works makes a share undefined. The PI set is fixed across categories within outcome but can differ across outcomes. Equal-PI within-state shares are not pooled participation percentages. Repository tables retain complete profiles and shared-work accounting. Pairing fixes identity, not time, career stage, institutional conditions, role capture or opportunity. Descriptive means/differences have no intervals; no adjusted content model is reinstated.

The complementary PPML count model includes PI/institution and institution-year fixed effects plus a proxy for years since first observed advisor/committee appearance, grouped as 0–2, 3–5, 6–8 and 9+. The proxy is not measured career age. No all-topic-output adjustment or offset enters this model. Its fitted populations and weighting differ from the descriptive comparison. The compatible direct degree comparison uses 1,272 shared PI/institution/year keys per degree, separate degree nuisance parameters and the original unequal funding windows. It is not obtained by dividing the headline fits.

Table S3.16 Fixed-effects count rate ratios

Degree target	Full RR	For-profit-excluded RR	Approx. pointwise 95% CI	Three-target-adjusted p
Research doctorate	1.278	1.278	1.158–1.410	.0074
Master's	1.784	1.784	1.362–2.337	.0220
Master's / doctorate (common fit)	1.388	1.388	0.338–5.705	1.000

Table S3.17 Fixed-effects count retained populations

Degree	Population	PI-year rows	PIs	Institutions	Funding switchers
Research doctorate	Primary panel	21,803	3,354	199	1,500
Research doctorate	Headline fit	10,809	1,713	166	747
Research doctorate	Direct common fit	1,272	229	56	103
Master's	Primary panel	8,565	1,129	138	647
Master's	Headline fit	2,247	425	96	210
Master's	Direct common fit	1,272	229	56	127

A reanalysis preserved models, populations and points. Covariance groups connect institutions through clear investigator links on all-topic works in retained degree populations, including excluded investigators and cross-degree paths. Within-group dependence is unrestricted; between-group independence remains assumed. A nonlinear profile-deletion variance approximation and working-Poisson Satterthwaite t reference adapt small-sample estimating-equation/fixed-effect methods (Mancl and DeRouen, 2001; Pustejovsky and Tipton, 2018, 2023), rather than exact linear-model CR2. Both fixed-effect dimensions are profiled; career terms remain estimated. The original unavailable-interval gate was reporting policy, not algebraic impossibility.

Doctorate/master's/direct targets have 112/67/37 groups, effective funding-information counts 6.06/4.36/3.96 and reference df 3.895/2.755/2.467. Pointwise t intervals accompany three-target Bonferroni adjustment; ownership-excluded inputs are identical. Direct comparison is imprecise, not equivalence. Broader grouping was selected post hoc after narrower groupings undercovered. Full-refit calibration used 2,400 datasets: four fixed-design scenarios × null/fitted slopes ×300. Observed pointwise 95% coverage was 96–100% and three-target simultaneous coverage 98.3–100%. This supports an approximate calculation only under examined designs, not arbitrary dependence or measurement/linkage uncertainty. Data notes retain scenarios and diagnostic alternatives.

S3.4 Exploratory opioid-topic comparison

Outcome and observed populations

The opioid comparison examines completions through 2023, the last observed year. The 2018 NIH HEAL launch supplies historical context; the first major award round was announced in September 2019 (National Institutes of Health, 2018, 2019a). Identified NIDA connections do not identify HEAL awards or project-specific opioid funding. The outcome is an observed opioid label in the adopted codebook field S4 substance-label union across in-scope aims, not C8 data type. It includes heroin, fentanyl, prescription opioids, opioid use disorder and supported opioid-receptor-targeted compounds, including endogenous-system research. This is broader than opioid misuse alone. Each work counts once, including multi-substance works.

The whole-corpus trend retains all 13,227 works, including 2,047 with an observed opioid label. Labels are incomplete for 353 works, nine of which are known opioid positives and remain included. An

incompletely coded work without an observed opioid label is not a verified negative. The existing measurement evaluation does not establish period-specific opioid coding accuracy.

Table S3.18 Annual whole-corpus opioid counts and label completeness

Year	All works	Opioid works	Opioid share %	Incomplete labels n	Known opioid positives among incomplete labels
2010	709	59	8.3	22	0
2011	711	53	7.5	32	0
2012	828	76	9.2	21	0
2013	849	87	10.2	15	1
2014	920	75	8.2	21	0
2015	900	96	10.7	29	1
2016	899	94	10.5	19	1
2017	960	125	13.0	31	0
2018	992	151	15.2	36	1
2019	1,092	226	20.7	28	2
2020	1,085	262	24.1	18	0
2021	1,087	243	22.4	26	2
2022	1,104	263	23.8	26	1
2023	1,091	237	21.7	29	0

The full descriptive pair contains 3,161 NIDA-connected works, including dual NIDA/NIAAA connections, and 9,160 works with no identified connection to either institute. The 906 NIAAA-only works are excluded from this pair. Annual descriptions retain works lacking adjustment metadata. The primary baseline is 2014–2017. Counts are annualized for unequal-length periods, while shares pool numerators and denominators within each period. The 2020–2023 window is descriptive; only 2021–2023 and 2022–2023 enter the two primary adjusted tests.

Table S3.19 Annual opioid profiles in the NIDA comparison pair

Year	NIDA: opioid / works (%)	Comparison: opioid / works (%)	Excluded NIAAA-only works
2010	31 / 175 (17.7%)	28 / 489 (5.7%)	45
2011	23 / 173 (13.3%)	30 / 488 (6.1%)	50
2012	30 / 247 (12.1%)	43 / 521 (8.3%)	60
2013	47 / 240 (19.6%)	40 / 558 (7.2%)	51
2014	31 / 239 (13.0%)	43 / 619 (6.9%)	62
2015	26 / 211 (12.3%)	68 / 641 (10.6%)	48
2016	26 / 211 (12.3%)	66 / 638 (10.3%)	50

Year	NIDA: opioid / works (%)	Comparison: opioid / works (%)	Excluded NIAAA-only works
2017	38 / 222 (17.1%)	86 / 681 (12.6%)	57
2018	34 / 209 (16.3%)	112 / 713 (15.7%)	70
2019	50 / 234 (21.4%)	171 / 784 (21.8%)	74
2020	61 / 242 (25.2%)	197 / 772 (25.5%)	71
2021	66 / 244 (27.0%)	172 / 765 (22.5%)	78
2022	73 / 253 (28.9%)	185 / 752 (24.6%)	99
2023	87 / 261 (33.3%)	145 / 739 (19.6%)	91

Table S3.20 Fixed-baseline period profiles: full descriptive pair

Completion period	Group	Opioid / works	Opioid share %	All works per year	Opioid works per year
2014–2017	NIDA	121 / 883	13.7	220.75	30.25
2014–2017	Comparison	263 / 2,579	10.2	644.75	65.75
2020–2023	NIDA	287 / 1,000	28.7	250.00	71.75
2020–2023	Comparison	699 / 3,028	23.1	757.00	174.75
2021–2023	NIDA	226 / 758	29.8	252.67	75.33
2021–2023	Comparison	502 / 2,256	22.3	752.00	167.33
2022–2023	NIDA	160 / 514	31.1	257.00	80.00
2022–2023	Comparison	330 / 1,491	22.1	745.50	165.00

Observed opioid shares rose in both groups, with most whole-corpus growth occurring among works completed by 2020. Equal proportional growth from different baseline shares can produce a positive absolute differential; neither annualized counts nor share ratios are the adjusted estimand below. The later divergence in annual group shares was noted after inspection, and single-year values are imprecise. Full descriptive fractions cannot substitute for adjustment-eligible populations.

Adjusted comparisons and assumptions

Adjustment requires approved institution, assigned named degree, eligible broad field BF01–BF10 and resolved completion year. Unavailable fields and pending BF11 assignments are excluded. Repository Tables DEP3-E20–E22 retain sequential exclusions and matched descriptions; overlapping reasons are not additive. Samples contain 5,722 works/421 institutions and 4,814/395. Windows begin three/four years after launch, omitting launch/transition years. They are different completion cohorts, not repeated trainees/projects or measured award-to-completion intervals.

The linear probability model includes NIDA status, its interaction with the post period, and fixed effects for fine credential family, broad field, institution and completion year. Year effects absorb the post main effect. The interaction multiplied by 100 is an adjusted differential change in opioid share in percentage points. Institution-cluster CR1 covariance retains the actual design rank and uses a t reference with G−1

degrees of freedom (Cameron and Miller, 2015). Pointwise intervals and raw p values accompany Holm adjustment across exactly the two primary window tests (Holm, 1979); this does not cover all exploratory decisions. The analysis and windows were developed after examining trends and were not preregistered.

Table S3.21 Institutional overlap and four-cell comparison support

Window	All primary institutions	Both periods	Both groups	Both groups in both periods	Works in four-cell institutions
3	421	259	149	81	2,826 (49.4%)
4	395	235	138	72	2,278 (47.3%)

Window 3 denotes 2021–2023 and Window 4 denotes 2022–2023, each against 2014–2017. Institutional support sets overlap and must not be added. Only 49.4% and 47.3% of the samples occur in institutions supplying both connection groups in both periods. The broader primary additive design does not require all four cells at every institution.

Table S3.22 Primary adjusted differential changes in opioid share

Post period	Estimate pp	Pointwise 95% CI pp	Raw p	Holm p	Works / institutions
2021–2023	+2.90	−1.98 to 7.78	0.243288	0.271744	5,722 / 421
2022–2023	+4.31	−1.36 to 9.97	0.135872	0.271744	4,814 / 395

Figure S3.2 Opioid trends and completion windows



(A) Observed opioid shares for 3,161 NIDA-connected works (including dual connections) and 9,160 works without either institute connection, 2010–2023; 906 NIAAA-only works are excluded. Dashed line: 2018 HEAL launch. Adopted S4-label-union opioid positives, including incomplete-label known positives, count once. (B) NIDA-by-post linear-probability interactions ×100, with pointwise institution-cluster CR1 95% intervals and t(G−1). Baseline: 2014–2017; post: 2021–2023 (5,722 works/421 institutions) and 2022–2023 (4,814/395). Fixed effects cover fine credential, BF01–BF10, institution and completion year. Adjustment-eligible populations differ from A. Positive values denote larger absolute increases among NIDA-connected works; both intervals include zero (Table S3.22: raw/Holm p values). Connections do not identify HEAL awards, windows are not award-to-completion lags, and no causal funding effect is isolated. Observation ends in 2023 (S3.4; repository opioid tables).

Both pointwise intervals include zero and substantively larger positive differences. They do not establish equivalence, absence of an additional increase or a causal funding effect. Completion-year effects absorb increases common to both groups, so these models do not test whether HEAL-era funding contributed to their shared rise. Interpreting an interaction as differential change associated with funding would require a stable conditional NIDA–comparison gap absent that change; the annual pre-2018 diagnostic contradicts a constant-gap account.

Table S3.23 Pre-2018 adjusted NIDA-gap diagnostics

Diagnostic	Estimate pp	Pointwise 95% CI pp	Test statistic	Raw p	Works / institutions
Linear differential trend, 2010–2017	−0.77	−1.90 to 0.35	t(366)=−1.355	0.176401	5,076 / 367
2010–2013 versus 2014–2017 differential change	−2.99	−7.69 to 1.72	t(366)=−1.247	0.213160	5,076 / 367
Joint annual gap contrasts, 2010–2016 versus 2017	—	—	F(7,366)=2.7493	0.008559	5,076 / 367

The linear diagnostic is percentage points per year, the block diagnostic is percentage points, and the joint annual test has no single percentage-point estimate. All use the same eligible 2010–2017 pair and controls. Nonsignificant linear and block summaries do not establish parallel pretrends or negate the annual joint test. These separate diagnostics have no two-test Holm adjustment.

Institutional comparisons materially change results. Four-cell support gives −1.53/−3.56 points across windows; institution-specific post shifts give −1.56/−0.99. Imprecision precludes negative-effect claims. Repository Tables DEP3-E23–E25 retain all sixteen sensitivity rows and ten largest-institution deletions. Alternate-comparator raw p=.053059 remains above .05 and is not primary.

Group membership can change when an investigator becomes funded, and the comparison may receive unmatched or other NIH support, including NINDS-led HEAL pain funding. Stable credential, field and institution effects, comparable measurement and repository coverage, and sufficient independence across institutions remain assumptions. A historical PI-linked component contains about 54.5% of each adjusted sample, with size-effective component count near 3.3; its covariance sensitivity does not certify uncertainty. Institutional effects do not remove cross-institution dependence, and the intervals omit retrieval, coding, linkage and source-selection uncertainty.

Linear fitted values reach about −0.30. Literal negatives number 752/5,722 and 619/4,814; values below -1×10^{-10} number 650/5,722 and 618/4,814. Twelve nominally above-one values in the second fit are roundoff; none exceeds $1+1\times10^{-10}$. Repository Table DEP3-E27 retains both definitions/distributions. Predictions were not clipped or treated as calibrated individual probabilities.

Budget authority, obligations, research activity and completion occur at different times. Many projects began before HEAL awards; these windows capture at most early completions the awards could have influenced. Later award cohorts extend beyond 2023; this is not an award-linked cohort followed to graduation. Nonconstant earlier gaps, sensitivity to institutional comparisons, changing composition and absent HEAL-specific exposure prevent attribution. Neither positive estimates nor nonsignificant tests establish or exclude HEAL-associated change or predict later gains. Evaluation requires award-specific project-to-work links and post-2023 completions.

S4 Employment and role measurement

This section describes employment follow-up and the development and application of the Research and Clinical Role Measurement Instrument (RC-RMI). The instrument estimates separate, potentially overlapping research and clinical roles from occupation and industry codes when linked work-activity observations are unavailable. The results describe retained employment records and sensitivity to scoring assumptions; they do not directly measure individual duties.

S4.1 Employment observations and scoring bases

The sampling frame comprised 13,229 in-scope records representing 13,227 distinct works. We selected 325 records by stratified random sampling, jointly stratifying completion-year band (2010–2014, 2015–2019 and 2020–2023), degree category and broad field. The overall sampling fraction was 2.4567%; stratum allocations and inclusion-probability calculations are documented in the sampling companion. Selection probabilities and employment ascertainment are separate quantities.

The two authors followed employment using professional profiles (e.g., LinkedIn), institutional and employer pages, and other public sources. Sixteen selected records could not be located; failure to locate a record was not evidence of unemployment, although some of the 16 cases we suspected included unemployment. As this is an exploratory pilot, we did not count unemployment as a distinct category.

The retained analytical dataset contains 307 linked records, completed in October 2026. Each record retains one occupation–industry pair, coded using the 2018 SOC structure and six-digit 2022 NAICS categories. Sixty-five records indicated multiple current roles; the retained pair does not enumerate all concurrent jobs. Recorded identity-match uncertainty remains explicit. We use unweighted counts and percentages. Of 307 observations, 303 have RC-RMI scores and four remain unscored with unresolved applicability. These four are distinct from the instrument's insufficient-information value U, which occurs in no application record.

Table S4.1 Employment observation and scoring bases

Degree category	Observed n	Scored n	Unscored n	% of all records
Research doctorate (PhD/ScD and equivalent)	190	189	1	61.9
General master's	63	61	2	20.5
PsyD	24	24	0	7.8
Other named doctorate	17	17	0	5.5
Field-specific master's	13	12	1	4.2
Total	307	303	4	100.0

Counts are linked records, not verified distinct people. Two later-PhD overrides preserve the original linked MA metadata. Unscored observations remain in full observed denominators and are excluded only from explicitly labeled scored-base summaries.

S4.2 Instrument construction and scoring rules

Research and clinical functions are distinct from occupation and employment sector (Mathur et al., 2018). Research comprises knowledge creation, R&D and scientific or technical direction of that work; routine analysis, teaching and administration alone are insufficient (OECD, 2015). Clinical care includes prevention, assessment, treatment and direct or patient-specific indirect professional work for identifiable patients or clients. A clinical title, credentials, health-sector employment or research contact alone does not establish care.

Two human reviewers supplied ratings informed by official occupation descriptions, O*NET tasks and NAICS definitions. The protocol specified independent initial ratings followed by review of agreement and disagreements. Reviewed inputs cover 303 occupations and 68 industries, or 20,604 possible pairs. Initial ratings, adjudication and subsequent refinements are retained. We used O*NET task evidence from release 30.3.

Table S4.2 Role scores and operational meanings

Score	Publication label	Research meaning	Clinical care meaning
0	Least likely	Adequate occupation and setting evidence provides little affirmative indication of research	Adequate evidence provides little affirmative indication of direct or patient-specific care
1	Less likely	Nonresearch functions predominate, while incidental or uncommon research remains plausible	Nonclinical functions predominate, while uncommon care remains plausible
2	Indeterminate / mixed	Research and nonresearch functions coexist, conflict or cannot be distinguished	Clinical and nonclinical functions coexist, conflict or cannot be distinguished
3	More likely	Substantial research evidence, with remaining occupational or setting ambiguity	Considerable direct or patient-specific indirect care evidence, with nonclinical assignments still plausible
4	Most likely	Strong research-function evidence with a corroborating setting or an approved specific rule. Within our sample of 303 SOC codes and corresponding O*NET task evidence, “4s” are those “most likely” to be conducting research.	Care-defining occupation evidence with a corroborating care setting or an approved specific rule
U	Insufficient information	Information is insufficient, residual, unmapped or irreconcilable	Information is insufficient, residual, unmapped or irreconcilable

Scores 0–4 are ordered role judgments, not probabilities or work-time fractions; 2 means mixed/nondiscriminating evidence, U insufficient information without ordinal rank. Assignment and identity-match confidence differ. Context describes establishments: R0 is nonresearch-primary; R1 research-compatible but nonspecific; R2 recurring research alongside nonresearch functions; R3 research-primary. C0 is nonhealth/noncare-primary; C1 health-related noncare/compatible; C2 regular care alongside noncare; C3 care-primary. Context U is insufficient information. Context cannot create a role from occupation base 0. Repository Table DEP4-E02 defines both axes; university/hospital R2 overrides are judgments about heterogeneous settings.

Table S4.3 Frozen combination matrix on either axis

Occupation base	R0 or C0	R1 or C1	R2 or C2	R3 or C3	RU or CU
0	0	0	0	0	0
1	0	1	2	2	1
2	1	2	3	3	2
3	3	3	3	4	3
4	4	4	4	4	4
U	U	U	U	U	U

The fixed matrix is a rule lookup, not a fitted model. Its 60 entries cover six occupation-base values, five contexts and two axes. Base 0 and U remain unchanged; strong bases 3–4 are protected from contrary context. Missing context is not negative evidence. On the clinical axis, the corroboration gate is applied before the matrix.

Table S4.4 Clinical corroboration gate

Occupation care evidence	Meaning	Treatment of declared C2 or C3
Direct care	Apparent interaction with identifiable patients or clients in care delivery	Retain declared positive context
Patient-specific indirect	Responsibility for identifiable patients without clear direct interaction	Retain declared positive context
Mixed direct and indirect	Affirmative evidence of both forms of care responsibility	Retain declared positive context
No care function identified	Adequate evidence supports a nonclinical function	Replace positive context with C1
Unknown directness	Available evidence cannot determine care directness	Replace positive context with C1

C2/C3 requires affirmative direct or patient-specific indirect care evidence. C0/C1/CU pass unchanged; no-care or unknown directness reduces C2/C3 to C1. Reviewed no-care occupations require clinical base 0–1 and final score ≤1; inconsistencies fail checks and U is exempt. Directness is nominal. The scorer lookup and repository diagnostics enumerate possible pairs, not workers or independent duty observations.

Table S4.5 Joint role classification

Joint state	Rule	Interpretation
Research only	Research at least 3; clinical care below 3	Research meets the threshold; clinical care does not
Clinical care only	Clinical care at least 3; research below 3	Clinical care meets the threshold; research does not
Both research and clinical care	Both axes at least 3	Both functions meet the threshold
Other	Both axes below 3	Neither function meets the threshold
Unclassifiable	Either axis is U	Evidence cannot support a complete joint classification

The study-defined likely-role threshold is 3 on each axis. Other means neither axis reaches that threshold; it does not establish absence of either function or identify another career. U on either axis yields an Unclassifiable joint state. Known-axis marginals require stated bases. Unresolved or nonapplicable application records are not matrix scores; U, when present, remains in scored denominators.

S4.3 Development evidence and alternative specifications

Table S4.6 Agreement between original instrument reviewers

Rating field	Exact agreement, all pairs	Ordinal pairs	Within one, ordinal pairs	Weighted kappa (95% interval)
Occupation research	192/303 (63.4%)	287/303	283/287 (98.6%)	0.639 (0.571–0.696)
Occupation clinical care	256/303 (84.5%)	287/303	287/287 (100%)	0.888 (0.853–0.919)
Industry research context	54/68 (79.4%)	67/68	67/67 (100%)	0.680 (0.414–0.836)
Industry clinical context	64/68 (94.1%)	67/68	67/67 (100%)	0.961 (0.910–1.000)
Occupation care directness	272/303 (89.8%)	Not ordinal	Not applicable	Not applicable

Table S4.6 uses one original A–B pair per occupation/industry before adjudication. Exact agreement includes matching information-status values; ordinal statistics exclude U/nonordinal pairs (sixteen occupation pairs per axis, one industry pair per axis). Directness is nominal. Kappa intervals use 2,000 reviewed-unit percentile resamples, seed 20260907. Reproduced agreement measures instrument-input consistency, not workers' actual duties; reviewer expertise and detailed independence remain unestablished.

Research inputs were compared with 2023 SDR work-activity aggregates across 27 categories (NCSES, 2023, 2025). SDR covers research-doctorate holders, not linked duties here. Its aggregates informed refinements, making correspondence development evidence, not held-out validation. Repository Table

DEP4-E03 retains all ten score arms, crosswalk and weighting. Arms are occupation inputs, not final scores; survey uncertainty and mapping error are omitted. SDR professional services is not a specific clinical criterion. Neither axis has independent individual-duty validation.

Table S4.7 Research inputs under the three scoring specifications

Specification	Disputed industry research ratings	Establishment research overrides	Nursing faculty research base
Inclusive primary	Higher of the two ordinal ratings in each of 13 disputes	R2 for universities and three hospital industries	3
No hospital override	Higher of the two ordinal ratings in each of 13 disputes	Retain university R2; hospitals retain their dissent-selected contexts	3
Conservative	Lower of the two ordinal ratings in each of 13 disputes	No establishment overrides	2

Inclusive scoring is primary. It uses higher disputed ordinal research-context ratings, declared university and hospital R2 overrides, and research base 3 for postsecondary nursing instructors. Conservative scoring uses lower disputed ratings, omits those establishment overrides and assigns nursing instructors research base 2. The no-hospital-override alternative retains inclusive inputs except hospital-context promotion. All three use the same clinical axis, matrix, threshold and application records. The original declaration specified two alternatives; the no-hospital component sensitivity was declared later and is retained separately with the current three-specification application.

Thirteen disputes differ on both reviewers' ordinal R0–R3 contexts; RU disagreements are outside ordered dissent. R2 overrides use NAICS 611310/622110/622210/622310. Order is common occupation corrections, industry dissent, establishment overrides, nursing-faculty input, clinical gate, then matrix. Resolved primary inputs already contain these steps and are not resolved again. Specifications are not averaged; their range describes sensitivity to judgments, not intervals or calibrated bounds on duties.

S4.4 Occupations and roles by degree

Tables S4.8–S4.12 retain all five degree groups and full observed denominators. Named occupations require at least five observations within degree; all qualifying occupations are shown, smaller cells are pooled, and unavailable occupation is separate. Equal counts share competition rank and are ordered by SOC code. This reporting rule applies to these occupation tables, not universally to every aggregate or instrument pair. General master's and field-specific master's remain separate here.

Table S4.8 Research doctorates

Rank	SOC unit	Occupation or disposition	n	% within degree
1	19-1042.00	Medical Scientists, Except Epidemiologists	23	12.1
1	19-3033.00	Clinical and Counseling Psychologists	23	12.1
3	25-1071.00	Health Specialties Teachers, Postsecondary	22	11.6
4	11-9121.00	Natural Sciences Managers	11	5.8
4	25-1066.00	Psychology Teachers, Postsecondary	11	5.8

Rank	SOC unit	Occupation or disposition	n	% within degree
6	21-1014.00	Mental Health Counselors	9	4.7
7	11-9111.00	Medical and Health Services Managers	8	4.2
8	13-1111.00	Management Analysts	7	3.7
9	15-2051.00	Data Scientists	6	3.2
9	25-1113.00	Social Work Teachers, Postsecondary	6	3.2
11	19-1041.00	Epidemiologists	5	2.6
—	—	Other coded occupations with fewer than 5 observations each	58	30.5
—	—	Occupation unavailable	1	0.5
		Total	190	100.0

Table S4.9 General master's

Rank	SOC unit	Occupation or disposition	n	% within degree
1	19-3033.00	Clinical and Counseling Psychologists	9	14.3
2	19-1042.00	Medical Scientists, Except Epidemiologists	5	7.9
2	25-1071.00	Health Specialties Teachers, Postsecondary	5	7.9
—	—	Other coded occupations with fewer than 5 observations each	42	66.7
—	—	Occupation unavailable	2	3.2
		Total	63	100.0

Table S4.10 PsyD

Rank	SOC unit	Occupation or disposition	n	% within degree
1	19-3033.00	Clinical and Counseling Psychologists	15	62.5
—	—	Other coded occupations with fewer than 5 observations each	9	37.5
—	—	Occupation unavailable	0	0.0
		Total	24	100.0

Table S4.11 Other named doctorates

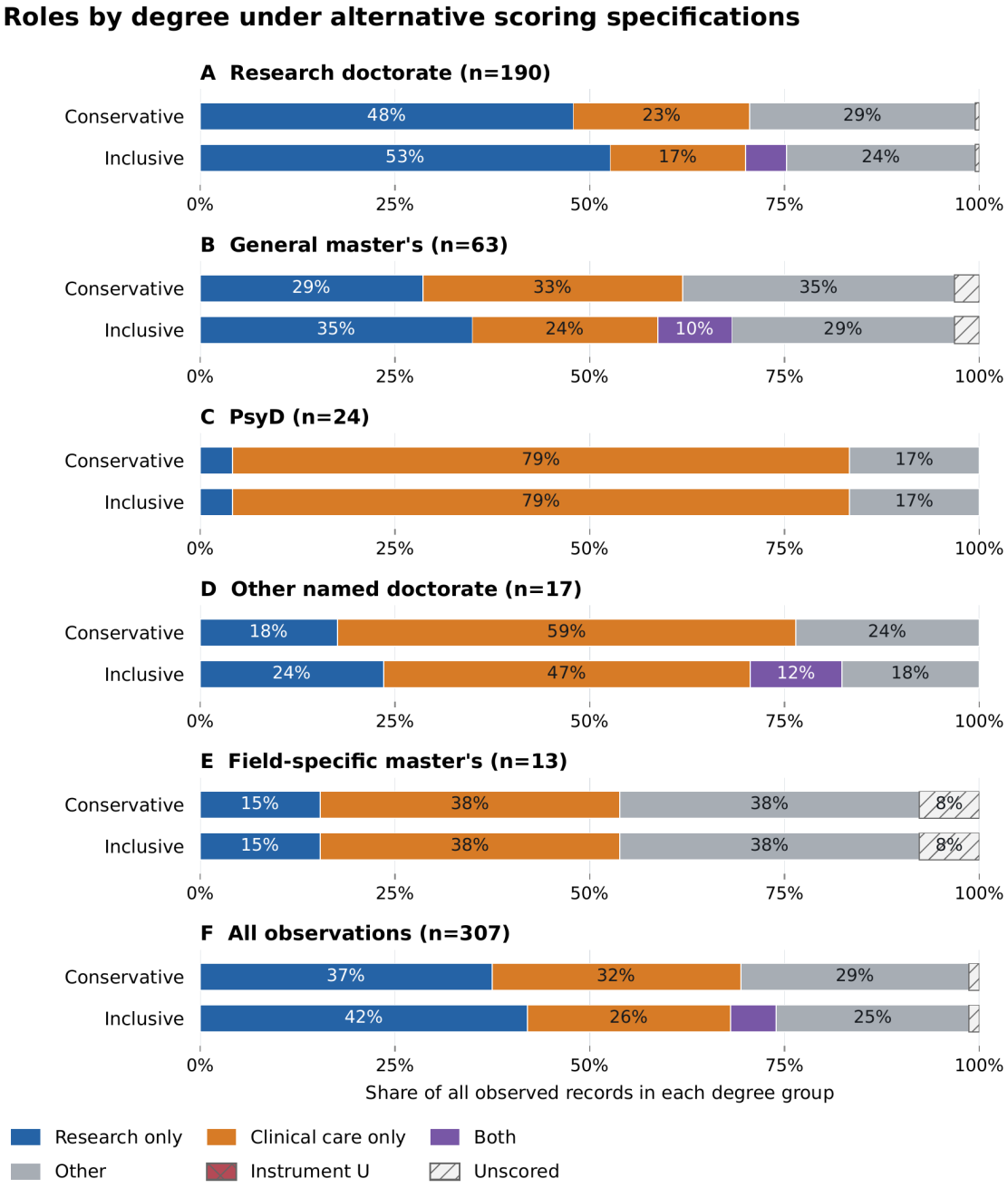
Rank	SOC unit	Occupation or disposition	n	% within degree
—	—	Other coded occupations with fewer than 5 observations each	17	100.0
—	—	Occupation unavailable	0	0.0
		Total	17	100.0

Table S4.12 Field-specific master's

Rank	SOC unit	Occupation or disposition	n	% within degree
—	—	Other coded occupations with fewer than 5 observations each	12	92.3
—	—	Occupation unavailable	1	7.7
		Total	13	100.0

Shares use every observed record in the stated degree, including unavailable occupation. No individual occupation reaches five observations among other named doctorates or field-specific master's records. Their coded occupations are therefore pooled rather than omitted.

Figure S4.1 Degree-specific roles under alternative specifications



Same observations and denominator in each pair. Clinical scores are unchanged.

Degree-specific roles under conservative and inclusive RC-RMI specifications. Each panel presents the same degree group's observations under both specifications, using its full observed denominator. The total panel includes all 307 observations. The likely-role threshold is 3 on each axis; Other means neither axis reaches that threshold. Instrument U (zero observations) is distinct from unscored applicability (four observations). Clinical scores are unchanged across specifications. Research differences represent alternative instrument judgments and provide a plausible specification range, not a statistical confidence interval or calibrated bounds on individual duties. Percentages printed in readable segments are rounded; complete plot data retain exact counts and unrounded shares.

Table S4.13 Research and clinical-care axis marginals

Degree	Observed n	Research conservative	Research no hospital override	Research inclusive	Clinical care all specifications
Research doctorate	190	91 (47.9)	103 (54.2)	110 (57.9)	43 (22.6)
General master's	63	18 (28.6)	21 (33.3)	28 (44.4)	21 (33.3)
PsyD	24	1 (4.2)	1 (4.2)	1 (4.2)	19 (79.2)
Other named doctorate	17	3 (17.6)	6 (35.3)	6 (35.3)	10 (58.8)
Field-specific master's	13	2 (15.4)	2 (15.4)	2 (15.4)	5 (38.5)
All records	307	115 (37.5)	133 (43.3)	147 (47.9)	98 (31.9)

Cells are n (% observed). Research includes Research only+Both; care includes Clinical care only+Both, so marginals overlap. Figure S4.1 pairs conservative/inclusive profiles; Table S4.13 adds no-hospital scoring. Clinical marginals are unchanged. Differences reflect alternative scoring of identical records, not temporal employment changes. Repository Table DEP4-E01 and scoring tables retain joint states, ordinal distributions, observed/scored bases and same-record transitions.

S4.5 Employment outside four-year colleges and universities

The outside subset requires known industry other than NAICS 611310, which covers colleges, universities and professional schools. Junior colleges and other education remain eligible; hospital jobs may retain academic affiliations. Unknown industry is separate. The subsets contain 120 research-doctorate records and 57 pooled general/field-specific master's records, all scored. The pooled master's observed base is 76, not the 63 general master's alone.

The repository outside-destination figures retain paired specifications and full outside-degree denominators. Inclusive Other shares are 32.5% for research doctorates and 38.6% for master's. Leading Other occupations include medical/health-services managers and management analysts for doctorates; medical/health-services managers, compliance officers and data scientists for master's, without individual sub-five counts. Other means neither role axis reaches threshold; Other industries is a separate destination residual.

Table S4.14 Outside-employment and academic-context sensitivities

Degree	Subset definition	n	Research conservative	Research inclusive	Clinical care all specifications	Other inclusive
Research doctorate	Outside four-year colleges and universities	120	42 (35.0)	49 (40.8)	38 (31.7)	39 (32.5)
Research doctorate	Recorded academic-context screen	82	31 (37.8)	31 (37.8)	26 (31.7)	25 (30.5)
Research doctorate	Explicit negative affiliation screen	80	29 (36.2)	29 (36.2)	26 (32.5)	25 (31.2)
Research doctorate	Outside, excluding junior colleges	119	42 (35.3)	49 (41.2)	38 (31.9)	38 (31.9)
Master's	Outside four-year colleges and universities	57	9 (15.8)	16 (28.1)	25 (43.9)	22 (38.6)
Master's	Recorded academic-context screen	41	6 (14.6)	8 (19.5)	18 (43.9)	17 (41.5)
Master's	Explicit negative affiliation screen	40	6 (15.0)	7 (17.5)	17 (42.5)	17 (42.5)
Master's	Outside, excluding junior colleges	56	8 (14.3)	15 (26.8)	25 (44.6)	22 (39.3)

Cells are n (% of the stated subset). Both academic-context screens require an explicitly negative visible academic appointment, no academic employer-name signal and industry other than 611310. The recorded-context screen excludes affirmative affiliation but permits blank/8; the explicit-negative screen additionally requires affiliation 0. Neither proves absence of academic connections. These screens retain 82/41 and 80/40 research-doctorate/master's records, respectively. The junior-college sensitivity also excludes NAICS 611210, retaining 119/56. General and field-specific master's are pooled in these comparisons.

Industry contributes to scoring; destination–role patterns are descriptive applications, not independent validation. Repository employment tables retain three-specification profiles, detailed industries, academic-context intersections and funding-role/Figure 6 data. The configuration has 15 detailed destinations and eight displayed groups. The optional objective–employment extension is outside this distribution.

S4.6 Exploratory research-role comparisons by employment destination and identified connection

Table S4.15 compares inferred research roles for research-doctorate employment observations with/without identified NIDA/NIAAA connections: within NAICS 611310 colleges/universities/professional schools, outside them, and within the main employment figure's eight outside groups. These destination groups were fixed before this comparison. Likely research is score ≥3, including joint clinical-care roles.

Under inclusive scoring, likely research roles were identified for 30/52 connected observations (57.7%) and 19/68 observations without an identified connection (27.9%) outside NAICS 611310, compared with 33/36 (91.7%) and 28/33 (84.8%) within it. The outside-minus-inside difference between these gaps was 22.9 percentage points (approximate 95% interval −0.5 to 46.4), so the comparison does not establish that the association is stronger outside colleges and universities. Among the eight outside groups, connected observations had higher shares in four, a lower share in hospitals, and equal shares in the other three; in health-practitioner offices and other health and social care, no observation in either group had a likely research role. Directions were unchanged under conservative and no-hospital-adjustment scoring. Reweighting both groups to the broad-sector distribution of all 189 known-industry research-doctorate observations reduced the difference from 25.1 to 18.6 percentage points under inclusive scoring (Panel B).

Table S4.15 Likely research roles by employment destination and identified NIDA/NIAAA connection, research doctorates

A. Likely research roles by destination

Destination	Inclusive scoring			Conservative scoring		
	Connected	No identified connection	Difference	Connected	No identified connection	Difference
All observations	63/88 (71.6)	47/102 (46.1)	25.5	52/88 (59.1)	39/102 (38.2)	20.9
Colleges, universities and professional schools (NAICS 611310)	33/36 (91.7)	28/33 (84.8)	6.8	26/36 (72.2)	23/33 (69.7)	2.5
Outside NAICS 611310, known industry	30/52 (57.7)	19/68 (27.9)	29.8	26/52 (50.0)	16/68 (23.5)	26.5
Scientific R&D services	10/10 (100.0)	6/9 (66.7)	33.3	10/10 (100.0)	6/9 (66.7)	33.3
Hospitals	5/8 (62.5)	6/6 (100.0)	−37.5	1/8 (12.5)	3/6 (50.0)	−37.5
Offices of health practitioners	0/5 (0.0)	0/18 (0.0)	0.0	0/5 (0.0)	0/18 (0.0)	0.0
Other health and social care	0/2 (0.0)	0/6 (0.0)	0.0	0/2 (0.0)	0/6 (0.0)	0.0
Government and public programs	5/8 (62.5)	3/12 (25.0)	37.5	5/8 (62.5)	3/12 (25.0)	37.5
Consulting and other professional services	1/4 (25.0)	1/6 (16.7)	8.3	1/4 (25.0)	1/6 (16.7)	8.3
Pharmaceutical, device and instrument manufacturing	5/5 (100.0)	1/1 (100.0)	0.0	5/5 (100.0)	1/1 (100.0)	0.0
Other industries	4/10 (40.0)	2/10 (20.0)	20.0	4/10 (40.0)	2/10 (20.0)	20.0

B. Gap contrast and sector standardization (percentage points)

Measure	Inclusive	Conservative	No hospital adjustment
Outside NAICS 611310 difference	29.8	26.5	26.5
NAICS 611310 difference	6.8	2.5	6.8
Outside minus NAICS 611310 (approximate 95% interval)	22.9 (−0.5 to 46.4)	23.9 (−4.1 to 52.0)	19.7 (−3.6 to 42.9)
Known-industry difference, crude	25.1	20.5	23.5
Known-industry difference, sector-standardized	18.6	13.2	14.6

Panel A cells are likely research roles/observations (%); differences are connected minus no identified connection, in percentage points. Destination groups follow the main-text employment figure; NAICS 611310 covers colleges, universities and professional schools, and outside rows require known industry. One research-doctorate observation without an identified connection has unknown industry and is unscored; it appears only in All observations. No-hospital-adjustment scoring matches inclusive scoring except for hospitals (1/8 vs. 3/6), the outside total (26/52 vs. 16/68) and All observations (59/88 vs. 44/102). In Panel B, the outside-minus-NAICS 611310 contrast uses an exploratory independent-record HC3 normal approximation; sector standardization reweights both groups to the observed broad-sector distribution of all 189 known-industry observations and does not adjust for field, institution or cohort. All comparisons are unweighted descriptions of retained observations. No identified connection does not establish absence of support, and industry contributes to RC-RMI scoring. Pointwise Wilson and Newcombe intervals, Fisher exact p values, broad sectors and detailed destinations are in data/destination_connection.csv.

A further grouping, formed after the destination comparison, pools Scientific Research and Development Services (NAICS 5417) with Pharmaceutical and Medicine Manufacturing (3254); Navigational, Measuring, Electromedical, and Control Instruments Manufacturing (3345); and Medical Equipment and Supplies Manufacturing (3391). Membership was determined solely by these NAICS industry codes, without conditioning on funding connection or role classification. The group contained 25 research-doctorate employment observations: 15 with an identified NIDA/NIAAA connection and ten without one. Likely research roles occurred in 15/15 connected observations and 7/10 observations without an identified connection under inclusive, conservative, and no-hospital-adjustment scoring; scientific R&D services contributed 10/10 and 6/9, and manufacturing 5/5 and 1/1 (data/industry_focus.csv).

These unweighted retained-observation comparisons have small cells and establish no funding effects. Industry contributes to scoring. Equal counts across specified alternatives indicate scoring consistency, not independent duty validation; excluding hospitals makes agreement with no-hospital-adjustment scoring add little information.

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